

AI / ML ENGINEER

Abdalla Nadir Abdalla

Melaka, Malaysia | +60 11-6810 0277 | abdullah130306@gmail.com

[linkedin.com/in/abdalla-nadir-a6841532b](https://www.linkedin.com/in/abdalla-nadir-a6841532b) | github.com/abdulla-cc | abdulla-cc.github.io/abdalla-nadir-portfolio

SUMMARY

Final-year AI student at Multimedia University focused on engineering automated data pipelines and applying ML/DL to real-world datasets. Shipped a live, self-updating NLP pipeline with weekly CI/CD automation and public deployment, and an end-to-end ETL system processing 1,470 records with zero manual intervention. Experienced in optimized SQL (CTEs, window functions) and production-style ML workflows. Seeking an AI or ML engineering internship.

TECHNICAL SKILLS

Languages & ML/AI: Python, SQL, C++, Machine Learning, Deep Learning, LSTM, XGBoost, Random Forest, Feature Engineering, Model Evaluation, NLP / Transformers

Data Engineering & Tools: ETL Pipeline Design, SQLite, Pandas, NumPy, TensorFlow/Keras, Scikit-learn, HuggingFace Transformers, Streamlit, GitHub Actions, Matplotlib, Power BI, Google Analytics

PROJECTS

AI-Powered Autonomous Supply Chain Optimization & Risk Management System (SCOMS) In Progress, Expected Mar 2027

Final Year Project, B.CS (AI), MMU · Python, TensorFlow/Keras, Scikit-learn, XGBoost, Streamlit

- Designed a three-module AI pipeline integrating LSTM demand forecasting, Random Forest supplier risk classification, and EOQ/ROP inventory optimization for retail supply chain decision support
- Architected LSTM forecasting model (2 stacked layers, 64/32 units, 30-day sliding window) for 7-30 day demand prediction; configured XGBoost benchmark across RMSE, MAE, and MAPE evaluation metrics
- Built Random Forest classifier (100 estimators) scoring suppliers as Low/Medium/High risk across five operational features: delivery delay rate, fulfillment rate, lead-time variance, defect rate, disruption history
- Implemented inventory module computing EOQ, Reorder Point, and Safety Stock dynamically from forecast output; surfaced via Streamlit dashboard with role-based recommendation flow for store managers
- Finalized system architecture and validated dataset strategy across 5-chapter FYP research phase, selecting a hybrid LSTM + XGBoost approach — benchmarked against 11 peer-reviewed supply chain optimization papers (2020-2026) to justify model selection over RNN and ARIMA baselines

YouTube AI Sentiment Tracker — Live NLP Pipeline & Dashboard

Jul 2026

Python, YouTube Data API v3, SQLite, HuggingFace Transformers, Streamlit, GitHub Actions

- Engineered an end-to-end NLP pipeline collecting 650+ YouTube comments via paginated YouTube Data API v3 calls, with SQLite storage enforcing deduplication through primary-key constraints and INSERT OR IGNORE logic
- Applied language detection to isolate English-language comments (89% of corpus), then scored sentiment using a pretrained RoBERTa transformer (cardiffnlp/twitter-roberta-base-sentiment) over rule-based alternatives for robustness to informal internet language
- Designed a confidence-weighted aggregation method (label direction × model confidence) to compute monthly sentiment trends across three AI tools, visualized in an interactive Streamlit/Plotly dashboard with reliability encoded via sample-size-driven marker sizing
- Automated the full pipeline — collection, language detection, scoring, and aggregation — via a GitHub Actions workflow scheduled weekly with authenticated repository write-back, achieving a fully self-updating production system
- Deployed the dashboard publicly on Streamlit Community Cloud; surfaced a validated finding that YouTube commenter sentiment skews negative (~3:1) across ChatGPT, Gemini, and Copilot, with Copilot trending most negative in recent months

XGBoost Diabetes Risk Models — Clinical & Behavioral Datasets

December 2025

Python, XGBoost, SMOTE, GridSearchCV, Scikit-learn

- Built two parallel binary classification pipelines predicting diabetes onset — 2,000-patient clinical dataset (Glucose, BMI, BloodPressure, Insulin, Age) and a 50/50 balanced 13-feature behavioral/socioeconomic dataset
- Engineered 2 clinical interaction features (BMI×Age, Glucose/Insulin ratio); applied median imputation across 5 columns (956 Insulin, 573 SkinThickness nulls) before training
- Resolved class imbalance through SMOTE oversampling applied exclusively to the training split, lifting minority-class F1 to 0.76 — ensuring the model remained clinically viable for detecting at-risk patients rather than defaulting to majority-class prediction
- Executed 5-fold GridSearchCV across 432 hyperparameter combinations — optimal config (max_depth=7, n_estimators=300, lr=0.05) achieved 82% test accuracy on minority class
- Behavioral pipeline used stratified splits and StandardScaler normalization, achieving 79.4% accuracy with balanced precision/recall (0.79/0.80) — validated via confusion matrices on all splits

EXPERIENCE

Committee Member, Activities Division

Mar 2025 – Mar 2026

Arabic Culture Society — Multimedia University (Part-time)

- Coordinated end-to-end planning and on-site execution of multiple student engagement events across a 12-month term, working within a cross-functional committee structure
- Managed event logistics — scheduling, vendor and venue coordination, internal communications, and post-event reviews — and contributed to outreach strategy to grow attendance

EDUCATION

Multimedia University, Melaka, Malaysia

Mar 2024 – Mar 2027

Bachelor of Computer Science (Artificial Intelligence) · GPA: 3.4 · Dean's List

Relevant Coursework: Machine Learning, Deep Learning, AI Fundamentals, Data Analytics, Database Systems, Data Structures & Algorithms, Computer Programming (C++), Software Engineering

CERTIFICATIONS & ADDITIONAL INFORMATION

HCIA-AI V4.0 — Huawei | CCNA: Introduction to Networks | Claude Code 101 — Anthropic | Microsoft Power BI for Beginners | Google Analytics Certification

Languages: Arabic (Native), English (Professional working proficiency)